

# Antreas Antoniou

Machine Learning Research Scientist & Engineer

Edinburgh, UK · [iam@antreas.io](mailto:iam@antreas.io) · +44 7516 162339  
[antreas.io](https://antreas.io) · [GitHub](#) · [Google Scholar](#)

I design and train learning systems, diagnose failures and optimise for deployment. I build the surrounding GPU hardware, cloud clusters, evaluation frameworks and applications. Research spans meta-learning, multimodal world models and memory.

## Model training & delivery

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### Axiotic AI — Chief Science Officer / Head of Research

November 2025–present

- **DAEDALUS multimodal world-model series — 0.5B, 1B, 4B, 8B, 16B, 32B and 72B parameters:** persistent memory and latent reasoning, with a self-improvement flywheel combining user interaction, autonomous agent testing and Byzantine validation methods researched at Axiotic.
- Trained the series' **72B model from scratch on approximately 1T tokens** in 2026, using 64 H100/B100 GPUs on GCP (PyTorch, Accelerate, FSDP); owned run design, training performance and execution.
- Designed and ran CHRONOS training experiments on richer temporal learning objectives, personally training the model ladder to **536M parameters and 5.5B tokens on a single RTX 4090**.
- **Adamantium:** designed and executed architecture/training campaigns spanning BPE transformers, byte-convolutional and hybrid embedding models; **5.8–9.4M parameters, approximately 10B teacher-token exposure per run across repeated corpus passes**. Developed screening/full-evaluation workflows and investigated benchmark and provenance failures.
- **GCP and experiment infrastructure:** built Olympus on GKE with Kueue, Spot workers, Workload Identity and immutable GCS storage; integrated agents and campaign control. Built portable experiment tooling for pinned-source execution, checkpoint/resume and W&B/Hugging Face tracking.
- **Home GPU cluster:** designed and built the hardware from scratch, then configured a **two-node, five-GPU k3s cluster (4x RTX 4090, 1x RTX 5090)** with Kueue scheduling for research campaigns.

### Pieces for Developers — Principal Research Scientist & Head of ML Research

March 2025–March 2026

- Conceived, designed, implemented and trained **20–80M-parameter language and embedding models** for on-device use. Analysed failures, iterated on the models and optimised the resulting systems through to working deployment.
- Developed retrieval and memory systems, evolutionary deep-research methods and large-context small models; connected model design with evaluation and device constraints. Confidential results omitted.

### Malted AI — Senior Machine Learning Research Scientist / Engineer

May 2024–February 2025

- Built experimentation frameworks and trained/adapted **Llama 3.1, Phi 3.5 mini and Qwen2** using LoRA, QLoRA and full fine-tuning, including small-data settings with **32–2,048 examples**.
- Built teacher-committee synthetic-data pipelines, agreement-based selection and **8B-to-sub-600M model distillation** experiments.
- Conceived and built an evolutionary prompt-optimisation system for an 8B model in one day. Results and recipes remain confidential.

# Research engineering & learning systems

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## University of Edinburgh — Research Associate; PhD research

Research Associate, 2021–2024 · PhD, 2017–2021

- Developed **MAML++**, **learned self-critique** and unsupervised few-shot methods: implementation, training and analysis of adaptation with limited labels; built *How to Train Your MAML*.
- Developed **Data Augmentation GANs** for transferable augmentation; research also spans relational reasoning and architecture search.
- Built **GATE/EEVEE** multimodal benchmark/evaluation tooling and **TALI** audio–language–image/video data pipelines. Built research-group Kubernetes infrastructure and authored **kubejobs**.

## Research conception & technical development — MLP and MSc projects

Conceived and pitched much of a multi-year student research portfolio; developed ideas with groups through architecture design, experiments, supplied code and training diagnosis.

- **World models:** model-based RL in CarRacing; DARLA/CoinRun disentangled representations and transfer under compute constraints.
- **Multimodal generation:** CLIP-conditioned AttnGAN, relational CycleGAN and low-data CLIP/StyleGAN2 translation; supplied neural modules and architecture guidance.
- **Data-efficient learning:** EvolutionAugment, adversarial augmentation and loss-driven curricula for class imbalance; authored augmentation code used by project researchers.

## Selected systems — independent projects

- **Cliphoard:** built a native Swift/macOS application deploying **Ogma embedding models** I trained at Axiotic for on-device semantic search via CoreML; implemented encrypted storage and model installation, and tested tokenizer and CoreML–PyTorch output parity.
- **HephaestusZero:** conceived, designed and built an **information-theoretic Byzantine** multi-agent runtime: uncertainty-scaled redundancy, diverse reasoning paths and adversarial verification. Engineered persistent, multi-resolution memory formation and agentic retrieval, combining standing context with query-specific exact, tag and BM25 search; selects relevant memories and their level of detail within a context budget.

## Additional experience

**Google, 2020–2021:** few-shot transfer under extreme domain shift. **Amazon, 2016:** speech-science internship, Echo. **LeRobot Edinburgh, 2025:** initiated the hackathon, supplied robotics/GPU hardware and mentored teams. Earlier embedded-hardware design/programming at Lancaster.

## Education & selected research

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**PhD, Machine Learning**, Edinburgh, 2017–2021 · **MScR, Data Science (Distinction)**, Edinburgh, 2016–2017 · **MSc, Data Science (Distinction)**, Lancaster, 2014–2015 · **BEng (Hons), Computer Systems Engineering**, Lancaster, 2011–2014.

**Selected publications:** *How to Train Your MAML* (ICLR 2019); *Self-Critique* (NeurIPS 2019); *Contrastive Meta-Learning* (ICLR 2023); *einspace* (NeurIPS 2024).

**Tools:** Python, PyTorch, TensorFlow, FSDP/Accelerate; Swift, CoreML; GCP/GKE, Docker, Kubernetes/k3s, Kueue; W&B, Hugging Face.